

ROMEO: Towards the Design of Robot with Haptic Mediation for Remote Conflict

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ABSTRACT

The utilization of robotic haptic stimuli in mediating remote conflicts has shown promising potential. Understanding how users figuratively interpret the metaphor of these stimuli in actual conflict scenarios is vital for the informed design of future mediation robots. The primary objective of this research is to delve into these interpretations and their implications. In this paper, we introduce a mediation robot capable of varying its temperature and softness to embody two roles: Avatar, representing one party in a conflict to express emotions and states, and Companion, serving as an emotional companion for the user. Through a study with 30 participants, we aimed to uncover and analyze the figurative explanations provided by users in response to the robot's haptic stimuli during conflict situations. Findings indicate that temperature changes were associated with emotional states like anger (hot) and fear (cold), while softness levels were linked to concepts of rigidity (hard) and flexibility (soft). This exploration allows us to discuss the potential role of robotic haptic mediation in remote conflict, thereby contributing valuable insights for the design of future mediation robots.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Computer systems organization** → **Robotic components**.

KEYWORDS

Social Robot, Haptic Interaction, Robot Mediation, Remote Conflict

ACM Reference Format:

Yijie Guo, Ruhan Wang, Yuan-Ling Feng, Yao Lu, Xuezhu Wang, Fumihide Tanaka, and Haipeng Mi. 2024. ROMEO: Towards the Design of Robot with Haptic Mediation for Remote Conflict. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3613905.3650781>

1 INTRODUCTION

Remote communication has increasingly become a commonplace method of interaction in daily life. However, existing research indicates that conflicts are more likely to arise in remote communications compared to in-person communications [1, 10, 20]. This is attributed to the difficulty of timely intervention by a third-party mediator in remote conflicts and the general lack of non-verbal cues, such as social signals, in communication devices like smartphones. Robots with physical interaction capabilities have the potential to serve as mediators in communication processes, offering immediate mediation suggestions and supplementing social cues [7, 17, 21]. While there have been discussions about the robots to provide mediation, these have predominantly focused on the robots' mediation dialogue design and its impact on conflicts, with little exploration into non-verbal interaction design [8, 21].

Recent studies have shown that haptic stimuli, as a form of non-verbal robot interaction, have the potential to support verbal mediation during remote conflict. For example, Noguchi's work revealed robots that shift their center of gravity can enhance the seriousness of the conveyed message in remote communication [13, 14]. Similarly, Embo introduces a wearable hand puppet robot designed to reduce verbal aggression among children by translating the intensity of aggressive speech into tactile discomfort [16]. Additionally, some research has systematically explored how different robotic mediator roles influence the design of the dimensions of its haptic mediation and revealed the extensive potential of temperature and softness [22]. However, the exact impact of these haptic stimuli in real conflict situations on users remains largely unexplored.

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CHI EA '24, May 11–16, 2024, Honolulu, HI, USA
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ACM ISBN 979-8-4007-0331-7/24/05
<https://doi.org/10.1145/3613905.3650781>

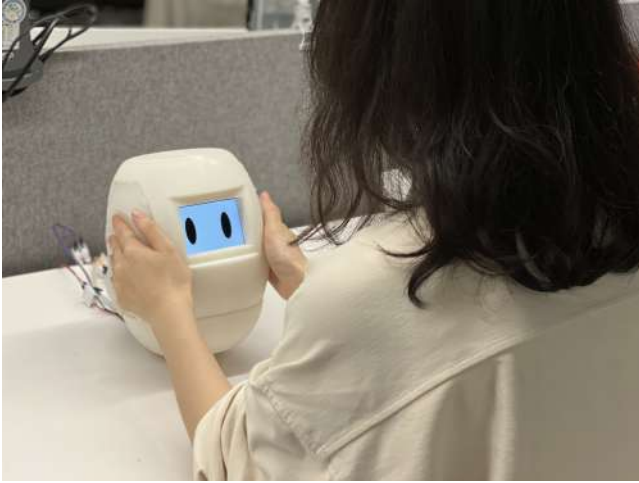


Figure 1: A robot mediator ROME0 which can provide haptic stimuli during the remote conflict.

To further explore how to design and interpret haptic mediation in robots that can be applied to remote conflict resolution, particularly focusing on temperature and softness stimuli, this research developed ROME0 (A RObot for MEdiation remOtely). ROME0 is a robot mediator capable of altering its temperature and softness, designed to allow users to discuss their understanding of these haptic mediations while engaging in everyday conflict scenarios. Through a study involving 30 participants, we explored users' figurative explanations of the haptic mediation provided by the robot in two mediation roles and discussed future design opportunities for remote conflict mediation robots. Therefore, the contributions of this paper are as follows:

- Documenting the design and development process of a mediation robot capable of varying in temperature and softness.
- Interpreting and discussing the potential role of haptic mediation from robots in remote conflicts, based on participants' revealed figurative explanations.

2 DESIGN EXPLORATION

Many social robots have already incorporated tactile elements in their design, primarily focusing on enhancing the haptic experience during interactions between the robot and users. However, there currently isn't a robot capable of providing timely tactile changes to mediate conflicts for users. To better explore user preferences for haptic mediation by robots and their potential interpretations to support the design of future mediation robots, we designed and developed ROME0 and conducted an exploratory study. In this section, we describe the design process of the ROME0 robot, from gaining inspiration to robot development and the design of haptic stimuli.

2.1 Design Consideration of ROME0

The previous study highlights a preference for desktop-sized robotic mediators, stemming from the belief that such robots' portability enhances remote communication[22]. Moreover, the research reveals

a general inclination towards employing temperature and softness in haptic rendering when robots assume various mediating roles. These haptic renderings are figuratively perceived as metaphors for emotional or attitudinal shifts. Drawing on the theory of conceptual metaphors in cognitive linguistics, some research indicates that users can vividly interpret a robot's temperature expressions through their mental models[15, 23]. For instance, Kövecses argues that anger can be metaphorically conceptualized as "hot fluid in a container", joy is associated with "warm" and fear is correlated with "cold"[9]. Additionally, the softness of a robot could be connected with figurative explanations. For instance, rigid is associated with "hard" and flexible is associated with "soft"[25]. However, the practical application and design of these haptic metaphors in robotic mediators require further exploration. In this study, we focus on understanding the metaphorical implications of softness and temperature renderings in various mediation roles. Therefore, our primary objective is to develop a robot capable of providing these haptic stimuli and embodying different mediating roles to support user share their understanding.

2.2 Development of the Robot ROME0

Inspired by the above considerations, we developed the robot ROME0. ROME0 is composed of three modules including softness adjusting, temperature adjusting, and display(fig. 2. A B). First, in the softness adjusting modules, two airbags (Polyethene, 130mm100mm) are connected to two micro air pumps (Kamoer KLC2B). These pumps can adjust the internal air pressure of the airbag by inflation and deflation, thus producing different softness stimulation. Secondly, in the temperature adjusting modules, four Peltier Cells are applied behind two pieces of silicone skin (Ecoflex 0030) to provide temperature variation. The silicone surface is designed with grooves so that the user can easily grip it. Both softness and temperature-adjusting modules are controlled via the PWM signal generated by an Arduino Uno. Finally, in the display module, a smartphone was installed as a robot face. It can display a neutral facial expression of ROME0 to help the participants understand the concept of the "robotic mediator" instead of some other devices.

2.3 Haptic Stimuli Design

Users were inclined to be mediated by temperature and softness variation while they were in the remote conflict. To investigate their understanding of metaphorical implications, here we designed haptic stimuli for ROME0.

Temperature. To understand the figurative explanations of thermal expression from robots in remote conflict, we designed temperature stimuli of ROME0 as shown in fig. 2. C. Referring to some related works on temperature stimuli models [15, 23], ROME0 would provide three basic temperature stimuli (hot: 35°C, neutral: 27.5°C, cold 20°C) as a reference for getting users' further explanation.

Softness. We designed ROME0's softness stimuli as shown in fig. 2. C. We conduct a pilot study with ROME0 to determine the design of its softness stimuli. In total, four people are asked to press ROME0 that had been inflated for 2, 4, 6, 8, or 10 seconds by the air pump operating at 2.4 l/min. Based on their feedback, we consider approximately four seconds of inflation as neutral and

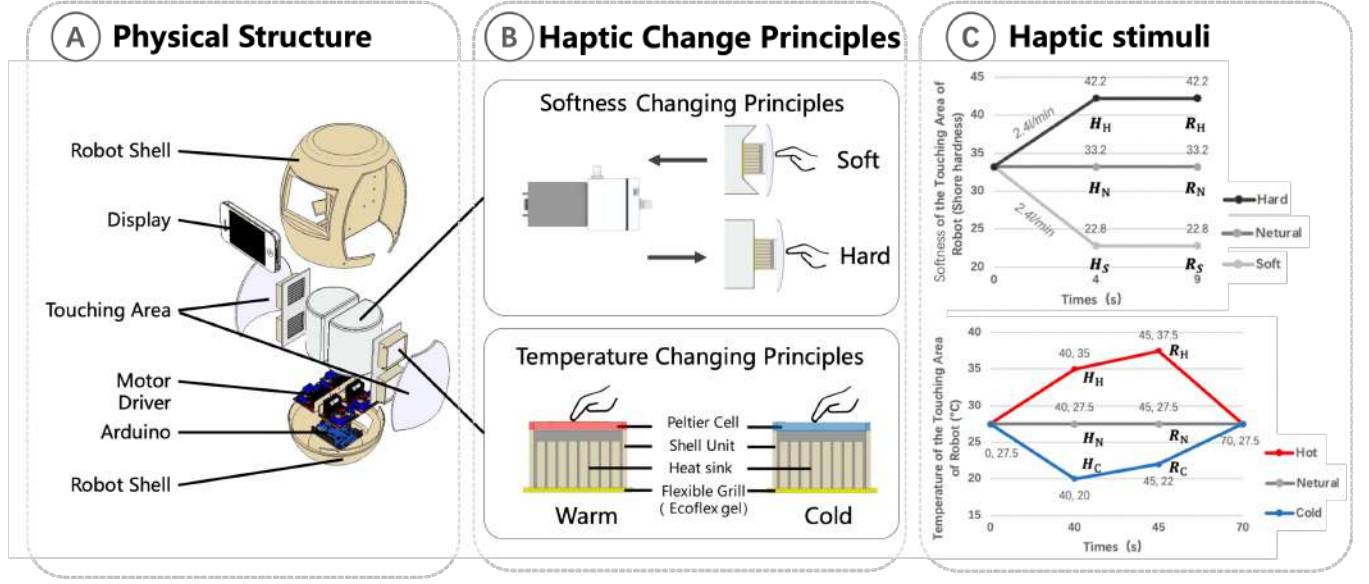


Figure 2: Design and development of ROMEO. (A. Explosion diagram of ROMEO's physical structure. B. Detailed principles for its haptic stimuli. C. Design details of haptic stimuli.)

eight seconds as hard. Each state was measured five times using the shore durometer LX-C-2. The mean values of the results were 42.2(hard), 33.2(neutral), and 22.8(soft). Thereby, ROMEO would provide these three basic softness stimuli as a reference for getting users' further explanations. Since different speeds of haptic stimuli changing may be perceived as different levels of arousal [23], as an exploratory study here we only focused on how the participants would figuratively explain temperature and softness level. Thus, when presenting the haptic stimuli, we asked the participants to feel the stimuli in four seconds. In fig. 2. C, H represents that the user holds the robot and R represents that the user releases. Due to technical limitations, small fluctuations in temperature changes occur during the holding process (fig. 2. C).

3 EXPLORATORY STUDY

In this research, we conducted an exploratory study to investigate the metaphorical implications of softness and temperature renderings in various mediation roles. Following the previous work, we set the roles of the robotic mediator as Avatar (representing one party in a conflict to express emotions and states) and Companion (serving as an emotional companion for the user). The participants were divided into two groups for each robotic mediator, and they were asked to write their subjective responses to a cartoon depicting a remote conflict and select a combination of haptic stimuli along with explaining their understanding of these stimuli based on the experience.

3.1 Context Scenario

In the following exploratory study, we randomly provided participants with seven cartoons depicting remote conflicts to create conflict scenarios. We drew inspiration from the P-F (Picture Frustration) study developed by Rosenzweig[18, 19], commonly

regarded as a semi-projective technique to assess participants' reactions to cartoons that depict various conflict scenarios [2]. To mitigate contextual bias, we endeavored to design diverse P-F study cartoons encompassing a wide range of conflict scenarios. Referring to a study [24] involving 890 Chinese university students on daily conflict categories, we categorized the themes of the cartoons into seven types: institutional structure, habit differences, cognitive differences, emotional attitudes, violation experiences, benefit competition, and communication barriers. Fig. 3 illustrates one of the scenarios from our P-F study cartoons. In each cartoon, the male character on the left says something that either expresses his frustration or frustrates the female character. All cartoons intentionally omit facial expressions, personality traits, and detailed contexts. Participants were instructed to assume the role of the character on the right and to complete the dialogue with their responses. The gender of the characters in the cartoons was altered to match that of the participants, aiding them in better relating to the character. We believe that participants' additions to the dialogue can, to an extent, reflect their expectations for mediation, which will provide reference and support for subsequent analyses.

3.2 Participants

We recruited 30 participants (15 males; 15 females; age: 19–29, $M = 24.57$, $SD = 2.67$) for this study. Participants were recruited from a university in China which were all Chinese speakers. The whole experiment was conducted in Chinese. Each participant is compensated with \$14. During the study, all participants provided informed consent.



Figure 3: Example of the P-F study cartoon we designed.

3.3 Procedure

Participants were divided into two equal groups to respectively experience the ROMEO robot as an Avatar character and as a Companion character. During the experiment, participants sat side-by-side with the experimenter in a chair facing ROMEO. A research assistant sat nearby with a laptop to activate the haptic stimuli. First, participants are briefly informed about the overall objective of the study and are asked to complete informed consent. Next, the participants are introduced to the mediation role of the ROMEO, depending on their group. Then, the participant was asked to feel each temperature and softness variation individually from ROMEO. Subsequently, the experimenter presented the participant with the contents of seven cartoons (described in 3.1) in a random order. Participants were allowed to imagine all the missing information within the cartoon by themselves. We assumed that this would be more accurate to obtain the real responses of the participants. After completing each cartoon, the participants were asked to select a haptic mediation for ROMEO to which they were inclined. Then ROMEO would provide these haptic stimuli. During the process, the participants could correct their selection at any time. After this process, participants were asked to give a figurative explanation for their selections. Finally, they were given a new cartoon to complete all 7 conflict scenarios. During the experiment, the display of ROMEO keeps displaying a neutral facial expression.

After completing all the cartoons, we conducted a brief semi-structured interview with the participants. During the interview, we discussed their opinions on the mediation style of ROMEO that they have experienced, and their suggestions for future robot haptic stimuli design.

4 RESULTS AND FINDINGS

The figurative explanations about haptic stimuli from participants were transcribed and analyzed using open coding, a prevalent qualitative analysis method in Human-Computer Interaction (HCI) and Human-Robot Interaction (HRI) [3]. Then, we conducted qualitative analyses based on these results. Before the qualitative analysis, we checked whether each participant correctly understood the robot's mediation style and haptic stimuli. One participant in the companion mediator group was unable to perceive the robot as a mediator. One participant in the avatar mediator group did not correctly understand the robot's mediation style. In all of the subsequent analyses, the data from them were excluded. To obtain more

objective analysis results, two evaluators independently extracted and integrated the haptic stimuli mentioned by the participants and their interpretations, then compared their findings and discussed the inconsistencies until produced consensus conclusions.

4.1 Varied Haptic Stimuli Offer Different Mediation Effects

4.1.1 Temperature Stimuli Related to Emotional and Affective Responses. Our findings indicate that participants, irrespective of being in the Avatar or Companion group, largely interpret temperature stimuli as capable of expressing, conveying, and mediating emotions in conflicts. (fig. 4. A) In the Avatar group, many participants noted that a robot's warmth was perceived as an indicator of the other party's anger or urgency, while a cool sensation suggested calmness or seriousness. Conversely, in the Companion group, temperature changes were primarily seen as a means to soothe their own emotions, with warmth providing support or comfort and coolness hinting at the need to remain calm.

4.1.2 Softness Stimuli Associated with Attitudes and Stances. Unlike temperature stimuli, which were emphasized for emotional conveyance, softness was strongly linked to the communication of attitudes (in the Avatar group) and support of stances (in the Companion group). (fig. 4. A) For instance, P3 interpreted the robot's softening at the end of a conflict as a symbol of the other party's genuine softening of attitude, facilitating conflict resolution. P21 felt that a hardening robot indicated a firm stance, aiding in appropriate response formulation. P4, in a systemic conflict scenario, chose a neutral softness for the robot, reflecting a considered expression rather than an impulsive emotional attack, thus easing mutual attitude perceptions.

4.2 Different Robotic Mediator Roles Establish Distinct Mediation Relationships

4.2.1 Avatar Mediators Expected to Present Emotions or Attitudes Directly or Indirectly. For most participants in the Avatar group, temperature and softness were understood as directly informing them of the emotional state and attitude of the conflict party. (fig. 4. B) For example, P24 desired information about the other party's emotional state in remote conflicts, which he believed could be accurately reflected by the robot. He mentioned, "If the robot heats up, it indicates the other party is angry, prompting me to pay more attention and patiently resolve the issue." P21 also interpreted a hardening robot as an indication of a strong stance from the other party, which would enable him to respond more appropriately.

Additionally, some participants hoped for the robot's temperature changes to moderate the negative emotions expressed in the conflict, with the robot acting as an intermediary that partially enhances or modifies such emotional cues. (fig. 4. B) For instance, P6, in a p-f animation depicting a communication barrier, imagined a couple's argument and stated, "The other's words are too cold; I wish the robot could warm up, filling the coldness in their words with warmth."

4.2.2 Companion Mediators Provide Positive Support and Assist in Alleviating Negative Emotions. Robots in the Companion role were perceived as mediators aligned with the user's stance, thus

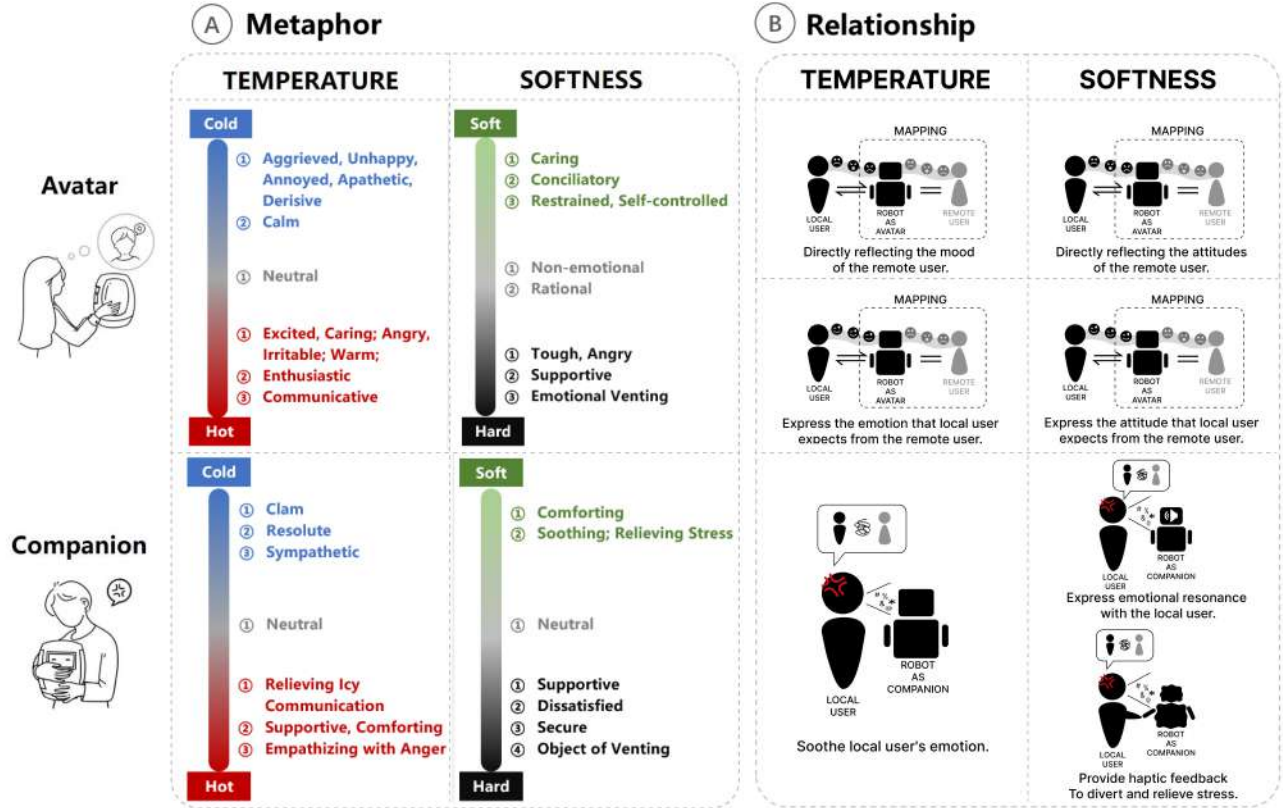


Figure 4: Results and findings of the exploration study. (A. Different dimensions of haptic stimulation provide different mediating effects. B. Robot mediators with different roles have different mediation relationships with users.)

participants valued the support and encouragement derived from temperature and softness stimuli. (fig. 4. B) One form of expressing support was the robot mirroring the participant's state, as noted by P9 in a p-f animation showcasing an emotional conflict. He said, "When the robot becomes hard and warm, it expresses anger and dissatisfaction similar to mine, resonating support." Another form of emotional support was through companionship, with most participants opting for a warm and soft robot, suggesting a metaphor of non-aggressiveness and affinity.

Notably, in the Companion group, the majority mentioned the potential of softness changes in aiding the venting of negative emotions (fig. 4. B), suggesting that softness variations could be key in alleviating psychological stress during mediation. P17 consistently chose a harder robot in various conflict scenarios, stating, "When the robot hardens, it gives a sense of resistance, allowing me to vent." P18 commented, "If the robot softens, I can press its body to relieve stress during the conflict." Analyzing these interpretations, we infer that regardless of whether hard or soft tactile sensations are more suitable for emotional venting, participants fundamentally seek a sense of exertion or feedback in tactile form, thereby transferring

some of their emotional pressure and achieving a decompression effect.

5 DISCUSSION AND FUTURE WORKS

5.1 Future Remote Robotic Mediator Design

During the interview of this study, participants also provided some suggestions for our haptic technology. For example, they wanted the hardness stimuli to present between a sofa-like touch and a steel-like touch, rather than a hardness variation on the silicone. For future work, designers may need to consider introducing more cutting-edge HCI haptic technologies (such as fluid interface[5, 11, 12], shape change technology[4, 26], etc.) to provide richer stimuli.

Users also presented some of their imaginations for robots. Most users mentioned that they would like to integrate robots into everyday objects. For our robot mediator ROMEO, users suggested that it can be integrated into household objects (e.g., cups, sofas) to provide peripheral interaction in remote conflicts; and anthropomorphic objects such as dolls to provide intuitive interaction. Some wearable devices (e.g., watches, U-shaped pillows) were also mentioned as potential robot designs.

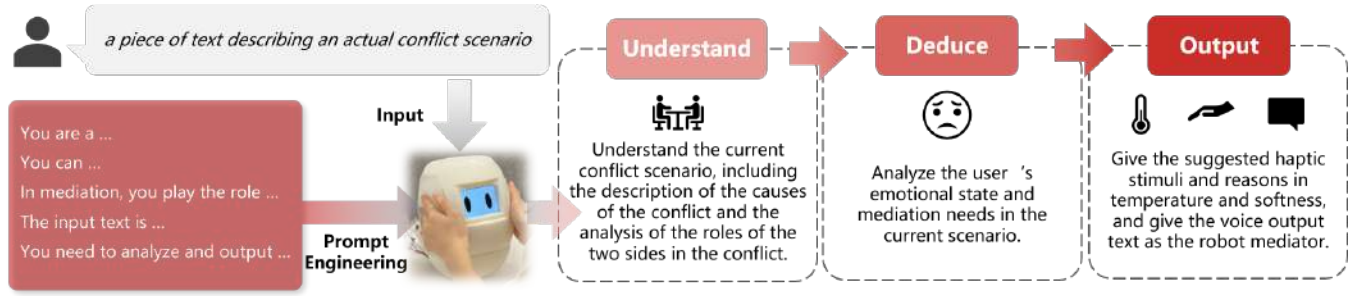


Figure 5: An example of mediation using a large language model to drive ROME.

5.2 Contextual and Cultural Factors in This Study

The conflict scenarios we presented during the study, which drew upon common conflict types in the lives of Chinese university students, were significantly shaped by cultural context and did not encompass all conflict situations encountered in daily life.

Also, in the aspect of explaining haptic mediation through metaphors, our exploration was primarily based on the insights of participants with native Chinese linguistic backgrounds. This cultural context might impact the interpretation of certain metaphors. In Chinese, the inherent meanings of characters that make up words can influence users' haptic associations with these words. For example, in the Chinese adjectives *rè qíng* (enthusiasm), *lěng jìng* (calm), *qiáng yìng* (tough), and *ruǎn ruò* (weak), the characters *rè*, *lěng*, *yìng*, and *ruǎn* correspond to the concepts of hot, cold, hard, and soft, respectively. Such metaphors could be subject to modification in cross-cultural contexts.

5.3 Importing LLM into the Design of Haptic Mediation

While our study primarily focused on text-based conflicts, which are asynchronous in nature, the principles and insights gained can be adapted to synchronous forms of communication. For instance, the haptic feedback mechanisms we explored, like temperature and softness changes, can be integrated into real-time communication scenarios to convey emotions and support mediation. The advancement of AI-mediated Communication (AIMC) technologies has enabled robots to accurately employ these mechanisms[6]. Particularly noteworthy is the recent maturation of Large Language Models (LLMs), such as OpenAI's GPT-4V, whose exceptional semantic comprehension and generation capabilities in natural language show promise in bridging the gap from the design of haptic mediation to the generation of personalized mediation in various conflict scenarios. We have experimented with optimizing the LLM's ability to manage haptic mediation provided by ROME through prompt engineering techniques, as illustrated in fig. 5.

By leveraging the GPT-4V model, we endowed the robot with preliminary capabilities to perceive and mediate a range of conflicts through dialogue. Subsequently, using prompts, the robot is positioned in different roles of remote conflict mediation. The conclusions drawn from our exploratory study are incorporated into

these prompts, providing a reasoned basis for the LLM to offer haptic stimuli in mediation. In practical interactions, the robot analyzes a user's descriptive speech about a conflict scenario in real-time or asynchronously, as well as their likely emotional state at the time, thereby inferring their mediation needs. Following this, in line with our study's discussion on users' metaphorical perception of haptic stimuli, the robot offers mediation in terms of temperature and softness. This approach facilitates non-verbal interaction, complementing the robot's existing dialogue mediation capabilities.

In the future, we aim to further augment the haptic mediation capabilities of ROME. This includes developing a more profound understanding of complex conflict scenarios and user individuality, innovatively interpreting and transforming haptic metaphors, and managing interaction hardware across multiple modalities. Our goal is to rigorously validate the positive impact of utilizing haptic stimuli in robots for remote conflict resolution. Such advancements are anticipated to contribute to the transition from AI-Mediated Communication (AI-MC) to Robot-Mediated Communication (R-MC).

6 CONCLUSION

This work delved into how users figuratively interpret haptic mediation imposed by robots in actual remote conflict scenarios. Specifically, our study aimed to explore and elucidate the application of temperature and softness as haptic stimuli in robotic mediation for remote conflicts. To address this, we developed ROME, a robot capable of altering its temperature and softness. Using a semi-projective technique, we assessed the responses of 30 participants to remote conflict scenarios presented through comics. Subsequently, we exposed them to haptic stimuli from the mediation robot and investigated their interpretations of haptic mediation. Our analysis, conducted under two robotic mediation roles, revealed that temperature stimuli are primarily associated with emotional and affective responses, while softness stimuli are linked to users' perceptions of attitudes and stances. Moreover, these figurative interpretations are influenced by the varying roles of the robotic mediator. We believe these insights can significantly contribute to a deeper understanding of the potential role of robotic haptic mediation design in remote conflicts.

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